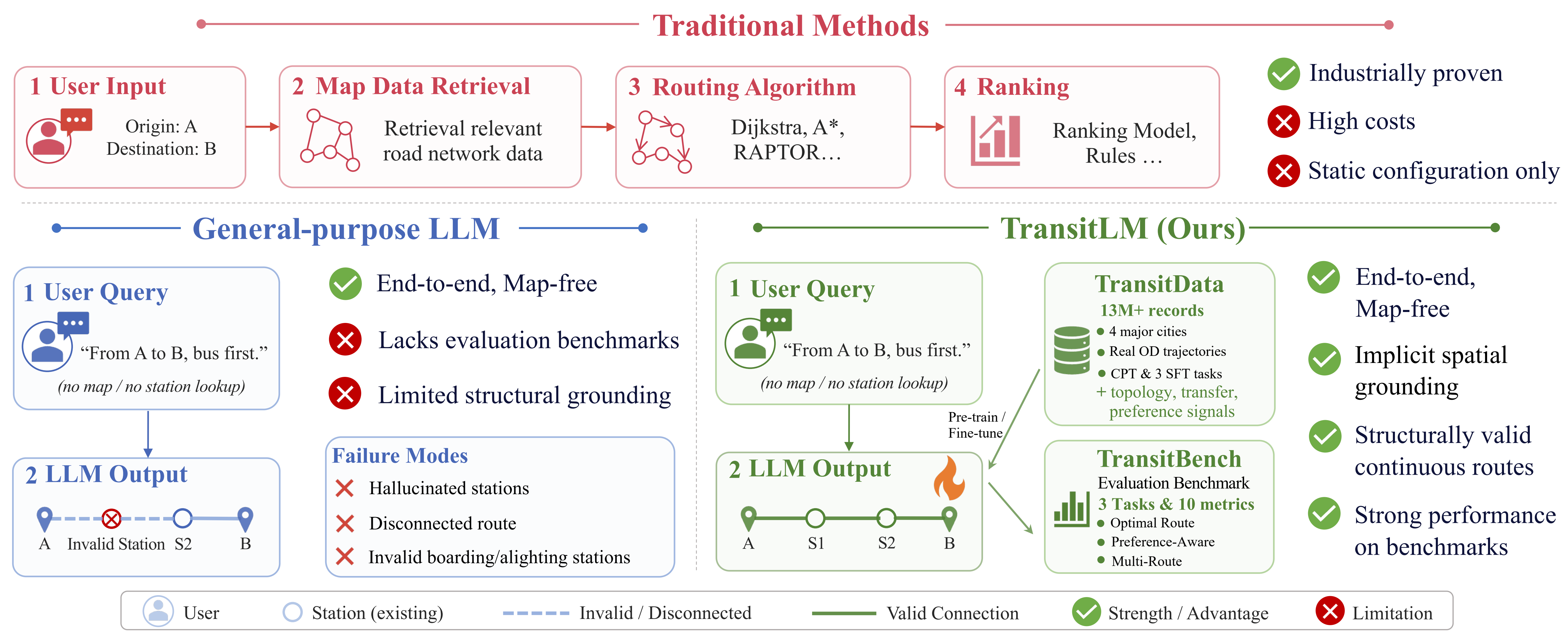

Traditional Methods
1 User Input
Origin: A
Destination: B
2 Map Data Retrieval
Retrieval relevant road network data
3 Routing Algorithm
Dijkstra, A*, RAPTOR…
4 Ranking
Ranking Model, Rules …
Industrially proven
High costs
Static configuration only
General-purpose LLM
1 User Query
"From A to B, bus first."
(no map / no station lookup)
2 LLM Output
A
Invalid Station
S2
B
End-to-end, Map-free
Lacks evaluation benchmarks
Limited structural grounding
Failure Modes
Hallucinated stations
Disconnected route
Invalid boarding/alighting stations
TransitLM (Ours)
1 User Query
"From A to B, bus first."
(no map / no station lookup)
2 LLM Output
A
S1
S2
B
Pre-train / Fine-tune
TransitData
13M+ records
4 major cities
Real OD trajectories
CPT & 3 SFT tasks
+ topology, transfer, preference signals
TransitBench
Evaluation Benchmark
3 Tasks & 10 metrics
Optimal Route
Preference-Aware
Multi-Route
End-to-end, Map-free
Implicit spatial grounding
Structurally valid continuous routes
Strong performance on benchmarks
User
Station (existing)
Invalid / Disconnected
Valid Connection
Strength / Advantage
Limitation